



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025 -26
MONDAY TEST

CLASS: IX
SUBJECT: MATHEMATICS

FULLMARKS: 40
DATE: 07/07/25

General Instructions:

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. [1×8 = 8]

i) The highest common factor of $24x^2y$ and $39xy^2$ is:

- (a) 13 (b) xy (c) $3xy$ (d) $312x^2y^2$

ii) The compound interest on ₹25000 after 2 years at 12% per annum will be:

- (a) ₹ 3630 (b) ₹6200 (c) ₹6300 (d) ₹6360

iii) The factors of $x^2 + xy + 8x + 8y$ are:

- (a) $(x + y)(x + 8)$ (b) $(2x + y)(x + 8)$ (c) $(x + 2y)(x + 8)$ (d) $(x + y)(2x + 8)$

iv) The difference between the amounts for ₹ 5000 after 1 year at 10% per annum compounded half-yearly and yearly will be:

- (a) ₹ 15.50 (b) ₹13.50 (c) ₹12.50 (d) ₹10.50

v) $(x + 8)(x - 10)$ in the expanded form of:

- (a) $x^2 - 8x - 80$ (b) $x^2 - 2x - 80$
(c) $x^2 + 2x + 80$ (d) $x^2 - 2x + 80$

vi) $\frac{1+\sqrt{2}}{3}$ is:

- (a) a rational number (b) an irrational number
(c) a natural number (d) an integer

vii) Statement 1: The commutative property is applicable to addition and subtraction both for rational numbers.

Statement 2: As per associative property, $A - (B - C) = (A - B) - C$. (A, B, C are rational numbers)

- (a) Statement 1 is only correct.
- (b) Statement 2 is only correct.
- (c) Statement 1 and statement 2 both are correct.
- (d) Statement 1 and statement 2 both are incorrect.

viii) Assertion (A): $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$

Reasoning (R): $4ab = (a + b)^2 - (a - b)^2$

- (a) Both (A) and (R) are true, and R is the correct reason for A.
- (b) Both (A) and (R) are true, and R is the incorrect reason for A.
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

SECTION: B

2. Factorise:

[3×3=9]

i) $x^2y^2 - xy - 72$

ii) $m^2 + \frac{1}{m^2} + 2 - 2m - \frac{2}{m}$

iii) $a^3 - 9b^3 + (a + b)^3$

3. convert 1.373737..... in rational form.

[3]

4. Mr. A keeps ₹ 4000 for 2 years at the rate of interest 8% per annum in a bank situated in his locality.

- i) Find the simple interest received after 2 years at the same rate p.a.
- ii) Find the compound interest after 2 years at the same rate p.a. compounded annually.
- iii) Calculate the difference between compound interest and simple interest received after 2 years.

[4]

5. Insert three rational numbers between $\frac{2}{7}$ and $\frac{3}{5}$.

[4]

6. If $x = \sqrt{5} + 2$, find the value of $x^2 - \frac{1}{x^2}$.

[4]

7. If $x + y = 8$ and $x - y = 2$, find the value of $2x^2 + 2y^2$

[4]

8. Calculate the amount on ₹16,000 in 3 years when the rates of interest are 5%, 10%, and 15% for successive years.

[4]